Dr. Yonatan Ashenafi

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Experience

Postdoctoral Fellow, Worcester Polytechnic Institute (WPI) Aug 2023 -
Worcester, MA

- Conducting research on modeling intracellular transport processes using experimental data from multi-university collaborations.
- Teaching Multivariable Calculus, Calculus I & II, and Ordinary Differential Equations.

Postdoctoral Fellow, University of Alberta Sep 2021 – May 2023
Edmonton, AB

- Researched motion mechanisms in diatoms (microscopic algae of ecological importance) through video-based data analysis.
- Taught Calculus I for Life Sciences and Calculus II for Physical Sciences.

Technical Fellow Intern, General Electric (GE) Research Oct – Dec 2020
GE Global Research Center, Niskayuna, NY

- Collaborated with the Probabilistic Design Team on experimental design and additive manufacturing projects.
- Developed Python programs for reinforcement learning tasks.

Teaching Assistant, Department of Mathematical Sciences 2016 – 2021
Rensselaer Polytechnic Institute (RPI), Troy, NY

- Led recitation sections, office hours, and exam review sessions for undergraduate courses.
- Assisted with instruction, grading, and student support for courses including Calculus I–III, Introduction to Differential Equations, Probability, Introduction to Data Mathematics, and Foundations of Applied Mathematics.

Research Assistant, Dordt College Jun – Aug 2014
Sioux Center, IA

- Constructed Gaussian mixture models from bacterial gene expression data using R.
- Gained experience in statistical data analysis and biological data modeling.

Tutor, Dordt College Library Sep 2013 – May 2016
Sioux Center, IA

- Tutored students in College Algebra, Calculus I–III, Linear Algebra, and Discrete Mathematics.
- Collaborated effectively with students, faculty, and staff to improve academic performance.

Education

Rensselaer Polytechnic Institute (RPI), Troy, NY
Ph.D. in Applied Mathematics, May 2021

Thesis: *Stochastic Hydrodynamics of Colonial Microswimmers*

Dordt University, Sioux Center, IA
B.A. in Mathematics, August 2012 – May 2016

Publications

- Ashenafi, Y. & Kramer, P. R. (2026). Stochastic Analysis of Kinesis and Taxis for Colonial Protozoa. *Bulletin of Mathematical Biology.*
- Ashenafi, Y. & Kramer, P. R. (2024). Statistical Mobility of Multicellular Colonies of Flagellated Swimming Cells. *Bulletin of Mathematical Biology*, 86, 125.
- Ashenafi, Y. (2022). Stability and Spatial Autocorrelations of Suspensions of Microswimmers with Heterogeneous Spin. *Physics of Fluids*, 34(12), 121908.
- Ashenafi, Y., Pandita, P., & Ghosh, S. (2022). Reinforcement Learning-Based Sequential Batch-Sampling for Bayesian Optimal Experimental Design. *Journal of Mechanical Design*, 144(9), 091705.
- Zheng, P., Kumadaki, K., Quek, C., Lim, Z. H., Ashenafi, Y., Yip, Z. T., Newby, J., Alverson, A. J., Jie, Y., & Jedd, G. (2023). Cooperative Motility, Force Generation and Mechanosensing in a Foraging Non-Photosynthetic Diatom. *Open Biology*, 13(10), 230148.
- Disselkoen, C., Greco, B., Cook, K., Koch, K., Lerebours, R., Viss, C., Cape, J., Held, E., Ashenafi, Y., Fischer, K., Acosta, A., Cunningham, M., Best, A. A., DeJongh, M., & Tintle, N. (2016). A Bayesian Framework for the Classification of Microbial Gene Activity States. *Frontiers in Microbiology*, 7, 1191.
- Ashenafi, Y. (Under review). Variational Dimension Lifting for Robust Tracking of Nonlinear Stochastic Dynamics.
- Ashenafi, Y. & Walcott, S. (In preparation). Simulation-Based Inference of Hidden Motor States in Intracellular Cargo Transport.

Service and Activities

- Online Instructor, Department of Information Technology and Artificial Intelligence, Shaggar Institute of Technology, Spring 2026
- Academic Consultant, Department of Information Technology and Artificial Intelligence, Shaggar Institute of Technology, 2025-present.
- Proposal Reviewer, National Science Foundation (NSF), 2025–present.
- Presenter, 2025 Canadian Mathematical Society Mathematics Education Meeting (Online).
- Judge, SIMIODE SCUDEM X 2025 (Differential Equations Modeling Challenge).
- Volunteer Grader, WPI Annual Math Meet, 2024.
- Presenter, Ethiopian Mathematics Professional Association Seminar, July 2025.

- Treasurer, Black Graduate Students Association, RPI (2020). Managed finances during the COVID-19 pandemic.
- Organizer, Mini-Symposium on Noise and Asymmetry in Microscopic Life, CAIMS 2022.
- Mathematical Modeler, Mathematical Problems in Industry (MPI) Workshop, Univ. of Vermont (2024) and RPI (2017).
- Participant, SIMIODE Spring 2024 Webinars.
- Presenter at Biophysical Society (2025),
- Presenter at Biology and Medicine through Mathematics (BAMM) conference (2024),
- Attended the Institute for Computational and Experimental Research in Mathematics (ICERM) Workshop on Interacting Particle Systems: Analysis, Control, Learning, and Computation (2024),
- Presenter at Society for Industrial and Applied Mathematics (SIAM) Life Sciences conference (2021),
- Presenter at Mathematical Biosciences Institute Worship (2020),
- Presenter at Univ. of Minnesota Multiscale Modeling Conference (2019),
- Judge, University of Alberta Undergraduate Research Symposium, 2022.

Awards

- Bill and Nancy Siegmann Applied Mathematical Modeling Prize, RPI (2021).
- DiPrima Summer Research Fellowship, RPI (2018).
- ASME CIE Advanced Modeling and Simulation Best Paper Award (2022).
- Viss Mathematics Scholarship (2016)

Technical Skills

Python (NumPy, Pandas, SciPy, TensorFlow, scikit-learn), MATLAB, C++, R, SQL

Professional Membership

Society for Industrial and Applied Mathematics (SIAM)